

Group Office Hours

February 25, 2021

ECON 301

Spring 2021



$$\text{Var}(\bar{Y} | X's) = \text{Var}\left(\frac{1}{n} \sum_i Y_i | X's\right) = \frac{1}{n^2} \text{Var}\left(\sum_i Y_i | X's\right)$$

$$= \frac{1}{n^2} \sum_i \text{Var}(Y_i | X's)$$

$$= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(Y_i | X_i)$$

$$= \frac{1}{n^2} \sum_{i=1}^n (\sigma^2)$$

$$= \frac{1}{n^2} n \sigma^2$$

$$= \sigma^2 / n$$

$$E[\hat{\beta}_1 | X_i] = \frac{1}{s_{xx}} \sum_i (x_i - \bar{x}) E[Y_i | X_i]$$

$$E[Y_i | X_i] = \frac{3x_i + 2}{6x_i + 3}$$

strong exogeneity

$$= \frac{1}{s_{xx}} \sum_i (x_i - \bar{x}) \left(\frac{3x_i + 2}{6x_i + 3} \right)$$

$$= \frac{1}{s_{xx}} \sum_{i=1}^3 (x_i - \bar{x}) \left(\frac{3x_i + 2}{6x_i + 3} \right)$$

$$s_{xx} = \sum_{i=1}^3 x_i^2 - (3) \bar{x}^2 = \sum_{i=1}^3 (x_i - \bar{x}) x_i$$

$$\bar{x} = ?$$

$$\frac{\sum_{i=1}^3 (x_i - \bar{x}) \left(\frac{3x_i + 2}{6x_i + 3} \right)}{\sum_{i=1}^3 (x_i - \bar{x}) x_i}$$

Incorrect!

$$\downarrow \sum_{i=1}^3 \frac{\cancel{(x_i - \bar{x})} \left(\frac{3x_i + 2}{6x_i + 3} \right)}{\cancel{(x_i - \bar{x})} x_i}$$

Review slides:

$$t = \frac{\bar{Y} - \mu_{y,0}}{SE(\bar{Y})}$$

$$; \quad \text{Var}(\bar{Y}) = \frac{\sigma^2}{n}$$

$$\widehat{\text{Var}}(\bar{Y}) = \frac{s^2}{n}$$

$$SE(\bar{Y}) = \frac{s}{\sqrt{n}}$$

$$s^2 = 0.08$$

$$\bar{Y}^{\text{act}} = 1.8$$

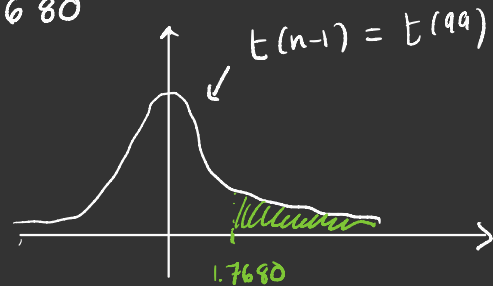
$$n = 100$$

$$t^{\text{act}} = \frac{1.8 - \mu_{y,0}}{\sqrt{\frac{0.08}{100}}} = \frac{1.8 - \mu_{y,0}}{0.02828}$$

$H_0: \mu_Y = 1.75$ vs $H_a: \mu_Y > 1.75$ one-sided right-hand test

$$t^{\text{act}} = \frac{1.8 - 1.75}{0.02828} = 1.7680$$

$$\begin{aligned} \text{P-value} &= \Pr(t(99) > t^{\text{act}}) \\ &= 0.04007 \end{aligned}$$

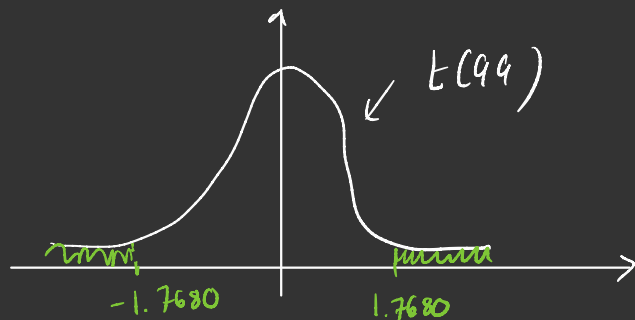


$$H_0: \mu_Y = 1.75 \quad \text{vs} \quad H_a: \mu_Y \neq 1.75$$

$$t^{\text{act}} = 1.7680$$

$$\begin{aligned} \Pr(|t(99)| > t^{\text{act}}) &= \Pr(t(99) > t^{\text{act}}) + \Pr(t(99) < -t^{\text{act}}) \\ &= 2 \Pr(t(99) > t^{\text{act}}) \\ &= 2 \Pr(t(99) < -t^{\text{act}}) \end{aligned}$$

$$\therefore p\text{-value} = 0.08014$$



$P_r (t(k) > t^{act})$: one-sided right-tailed

$P_r (t(k) < t^{act})$ one-sided left-tailed

$P_r (|t(k)| > t^{act})$ two-sided

where $k = \text{appropriate degrees of freedom}$

(usually for a standard simple linear reg, $k = n - 2$)

$Y = \text{wage}$

$X = \text{educ}$

$$E(\text{wage} | \text{educ}) = \beta_0 + \beta_1 \text{educ}$$

A2

+ A1 - A5

$$\frac{\partial E[\text{wage} | \text{educ}]}{\partial \text{educ}} = \beta_1$$

H_0 : The ceteris paribus effect
of education on wage
is zero

vs H_a : It is not

H_0 : education doesn't affect
your wage

vs H_a : It does

$H_0: \beta_1 = 0$ vs $H_a: \beta_1 \neq 0$

Types of Hypothesis tests

Two-sided: $H_0: \beta_1 = \beta_{1, \text{hyp}}$ vs $H_a: \beta_1 \neq \beta_{1, \text{hyp}}$

Right-sided: $H_0: \beta_1 = \beta_{1, \text{hyp}}$ vs $H_a: \beta_1 > \beta_{1, \text{hyp}}$

Left-sided: $H_0: \beta_1 = \beta_{1, \text{hyp}}$ vs $H_a: \beta_1 < \beta_{1, \text{hyp}}$